In [ ]:
```
############################ PART-1##########################
- Basic syntax

- Basic codes

- using eval basic codes

- Packages

- If-else

- try-exception

- Functions

- Loops

#################### PART-2##########################
1) Strings
2) List
3) Dictionary
4) File handling sessions
5) Tuple and sets

Statistics Theory  EDA with python Pandas numpy ma
```

Loops

- For loop
- While loop

In [ ]:
```
- Function will reuse the code

- tax payment of 10 members in your comapny

- def tax_pay():
-   tax payment = salary*tax_per/100
-   return tax_payment
```

In [1]:
```
for i in range(10):
    print(i)

# In python index start with zero
```

```
0
1
2
3
4
5
6
7
8
9
```

In [3]:
```python
#print(0)
#print(1)
#print(2)
#print(3)

for i in range(4):
    print(i)
```

```
0
1
2
3
```

In [4]:
```python
for i in range(5):
    print(i)
```

```
0
1
2
3
4
```

- Any loop we required 3 things
- Intial point: where you want to start
- Increment : How much you want increase
- Condition : when you want to stop
- In for loop these 3 in single line only

$$Pattern - 1$$

for i in range(stop)

- default start value =0
- default increment by =1
- increment direction is postive (+)ve
- last value=stop-1

In [5]:
```python
for i in range(5):
    print(i)

# i in range(5)
# intial point i=0
# incremant = 1
# stop point= 5-1
```

```
0
1
2
3
4
```

In [10]:
```
for i in range(20):
    print(i,end=' ')

# i=0
# pos i=1
# stop=20-1=19
```

```
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
```

In [9]:
```
print(0,end=' ')
print(1,end=' ')
print(2)

print(i,end=' ')
```

```
0 1 2
```

$Pattern - 2$

for i in range(start,stop)

- start value will be the value we provided at start position
- Direction postive
- Increment by 1
- last value= stop-1

In [11]:
```
for i in range(10,20):
    print(i,end=' ')

# start=10  stop=20-1=19  increment=1  direction=+ve
```

```
10 11 12 13 14 15 16 17 18 19
```

$Pattern - 3$

for i in range(start,stop,step)

- start value will be the value provided at start position
- step value
  - if step value has negative ex: -2, it is negative direction which means it is decreasing
  - if step value has postive ex: 2 , it has postive direction which means it is incremental
- stop value
  - if we have decremental last= stop+1
  - if we have incremental last = stop-1
- decremental will gives last=stop+1
- incremental will gives last=stop-1

In [12]:
```python
for i in range(2,20,2):
    print(i,end=' ')

# start=2
# step=+2  postive direction ==== increment
# last= stop-1 = 20-1=19

# 2  4  6  8  10 12 14 16 18 <19>
```

```
2 4 6 8 10 12 14 16 18
```

In [13]:
```python
for i in range(-1,-10,-1):
    print(i,end=' ')

# start=-1
# step=-1 negative direction
# last= stop+1= -10+1=-9

# -1      -9
```

```
-1-2-3-4-5-6-7-8-9
```

In [14]:
```python
for i in range(-1,10,-1):
    print(i,end=' ')

# start=-1
# step=-1  negative
# last= 10+1=11
```

In [15]:
```python
for i in range(5,-6,-1):
    print(i,end=' ')

# start=5
# step= -1  nega
# last= stop+1= -6+1=-5
```

```
5 4 3 2 1 0 -1 -2 -3 -4 -5
```

In [16]:
```python
for i in range(5,-6,1):
    print(i,end=' ')

# start=5
# step= +1  pos
# last= stop-1= -6-1=-7
```

In [ ]:
```
for i in range(2,20,2):
    print(i,end=' ')


for i in range(-1,-10,-1):
    print(i,end=' ')


for i in range(5,-6,-1):
    print(i,end=' ')


for i in range(5,-6,1):
    print(i,end=' ')
```

In [ ]:
```
range(3,25,3) # s=3  step=3 stop= 25-1 =24 p
range(3,25,-3) # s=3  step=-3  stop= 25+1=26 np
range(3,-25,3) # np
range(3,-25,-3) # p
range(-3,25,3) # p
range(-3,25,-3) # np
rang(-3,-25,-3) # p
```

In [ ]:
```
range(5) # stop
range(5,10) # start,stop
range(5,10,2) # start,stop,step
```

In [ ]:
```
range(<start>,<stop>)
```

In [17]:
```
for i in range(3,25,-3):
    print(i)

# No answer and No error
```

In [21]:
```
# WAP ask the user print the square of the number from 1 to 5
# 1*1=1
# 2*2=4
# 3*3=9
# 4*4=16
# 5*5=25

for i in range(1,6):
    print(f"{i}*{i} is {i*i}")
```

```
1*1 is 1
2*2 is 4
3*3 is 9
4*4 is 16
5*5 is 25
```

In [22]:
```
# Take start and stop value from keyboard
start=eval(input("enter the number:"))
stop=eval(input("enter the number:"))

for i in range(start,stop):
    print(f"{i}*{i} is {i*i}")
```

```
enter the number:1
enter the number:6
1*1 is 1
2*2 is 4
3*3 is 9
4*4 is 16
5*5 is 25
```

In [23]:
```
start=1
stop=6

for i in range(start,stop):
    print(f"{i}*{i} is {i*i}")
```

```
1*1 is 1
2*2 is 4
3*3 is 9
4*4 is 16
5*5 is 25
```

In [24]:
```

for i in range(eval(input("enter the number:")),
               eval(input("enter the number:"))):
    print(f"{i}*{i} is {i*i}")
```

```
enter the number:1
enter the number:6
1*1 is 1
2*2 is 4
3*3 is 9
4*4 is 16
5*5 is 25
```

In [ ]:
```
##########################################################
for i in range(1,6):
    print(f"{i}*{i} is {i*i}")

############################################################
start=1
stop=6
for i in range(start,stop):
    print(f"{i}*{i} is {i*i}")

###############################################################
start=eval(input("enter the number:"))
stop=eval(input("enter the number:"))

for i in range(start,stop):
    print(f"{i}*{i} is {i*i}")

###################################################################
for i in range(eval(input("enter the number:")),
               eval(input("enter the number:"))):
    print(f"{i}*{i} is {i*i}")
```

In [27]:
```
import random
start=random.randint(1,6) # 5
print(start)
stop=random.randint(7,10)  # 2
print(stop)



for i in range(start,stop):   # range(5,10)
    print(f"{i}*{i} is {i*i}")
```

```
4
10
4*4 is 16
5*5 is 25
6*6 is 36
7*7 is 49
8*8 is 64
9*9 is 81
```

In [29]:
```
# Function with arguments

def sqaure1():
    for i in range(1,6):
        print(f"{i}*{i} is {i*i}")

sqaure1()
```

```
1*1 is 1
2*2 is 4
3*3 is 9
4*4 is 16
5*5 is 25
```

In [30]:
```python
def sqaure2():
    start=eval(input())
    stop=eval(input())
    for i in range(start,stop):
        print(f"{i}*{i} is {i*i}")

sqaure2()
```

```
2
5
2*2 is 4
3*3 is 9
4*4 is 16
```

In [31]:
```python
start=eval(input())
stop=eval(input())

def sqaure3():
    for i in range(start,stop):
        print(f"{i}*{i} is {i*i}")

sqaure3()
```

```
4
8
4*4 is 16
5*5 is 25
6*6 is 36
7*7 is 49
```

In [34]:
```python
def sqaure4(s,last):
    for i in range(s,last):
        print(f"{i}*{i} is {i*i}")

sqaure4(4,8)
```

```
4*4 is 16
5*5 is 25
6*6 is 36
7*7 is 49
```

In [ ]:
```
#################################### Basic function with out arguments#######
def sqaure1():
    for i in range(1,6):
        print(f"{i}*{i} is {i*i}")

sqaure1()

################################ Local variable##############################
def sqaure2():
    start=eval(input())
    stop=eval(input())
    for i in range(start,stop):
        print(f"{i}*{i} is {i*i}")

sqaure2()

#################################### Global variable#########################
start=eval(input())
stop=eval(input())

def sqaure3():
    for i in range(start,stop):
        print(f"{i}*{i} is {i*i}")

sqaure3()


############################### With argument################################
def sqaure4(s,last):
    for i in range(s,last):
        print(f"{i}*{i} is {i*i}")

sqaure4(4,8)



def sqaure5(s=4,last=8):
    for i in range(s,last):
        print(f"{i}*{i} is {i*i}")

sqaure5()
```

In [37]:
```
10+20+30-6/6
```

Out[37]: 59.0

In [36]:
```
(10+20+30-5)/5    #BODMAS
```

Out[36]: 11.0

In [42]:
```
eval(input())
```

round(10/6)

Out[42]: 2

In [ ]:

```
Monday

I will upload a new assignment

pls ttend the labs

As of now how many assiments ? very bad
As of now how many assiments ?
exlain
```

In [47]:

```
val1=12345//10
```

In [45]:

```
n1=12345%10
```

Out[45]:

```
5
```

In [49]:

```
val1%10
```

Out[49]:

```
4
```

In [ ]:

```
# wap ask the user enter a number
# find whether the given number even or odd
# these above code you should implement on 5 numbers
# 5 times your code should display like below
#        enter the number
#        and you need to display the answer
```

In [3]:

```
num= eval(input("enter the number:"))
if num%2==0:
    print(f"{num} is an even number")
else:
    print(f"{num} is an odd number")
```

```
enter the number:19
19 is an odd number
```

In [8]:
```
for i in range(5):
    print("iteration:",i)
    num= eval(input("enter the number:"))
    if num%2==0:
        print(f"{num} is an even number")
    else:
        print(f"{num} is an odd number")
```

```
iteration: 0
enter the number:20
20 is an even number
iteration: 1
enter the number:30
30 is an even number
iteration: 2
enter the number:32
32 is an even number
iteration: 3
enter the number:33
33 is an odd number
iteration: 4
enter the number:34
34 is an even number
```

In [9]:
```
def even_odd():
    num= eval(input("enter the number:"))
    if num%2==0:
        print(f"{num} is an even number")
    else:
        print(f"{num} is an odd number")
```

In [10]:
```
for i in range(5):
    even_odd()
```

```
enter the number:1
1 is an odd number
enter the number:2
2 is an even number
enter the number:3
3 is an odd number
enter the number:4
4 is an even number
enter the number:5
5 is an odd number
```

In [ ]:
```
for i in range(5):
    print("iteration:",i)
    num= eval(input("enter the number:"))
    if num%2==0:
        print(f"{num} is an even number")
    else:
        print(f"{num} is an odd number")

for i in range(5):
    even_odd()
```

In [11]:
```
for i in range(5):
    number = eval(input("Enter the number"))
    if(number %2==0):
        print("Number is even")
    else:
        print("Number is odd")
```

```
Enter the number1
Number is odd
Enter the number2
Number is even
Enter the number3
Number is odd
Enter the number4
Number is even
Enter the number5
Number is odd
```

In [ ]:
```
#####################################################
def even_odd():
    num= eval(input("enter the number:"))
    if num%2==0:
        print(f"{num} is an even number")
    else:
        print(f"{num} is an odd number")

for i in range(5):
    even_odd()

###################################################

def evenodd():
    for i in range(5):
        number = eval(input("Enter the number"))
        if(number %2==0):
            print("Number is even")
        else:
            print("Number is odd")
evenodd()
```

In [ ]:
```
# wap ask the user to find whether the number is even or odd
# take the number randomly from 1 to 100
# this you need to implement 5 times
# which means every time it should take random number
# and display the answer even or odd
```

In [12]:
```
import random
for i in range(5):
    number = random.randint(1,100)
    if(number %2==0):
        print(f"{number} is even")
    else:
        print(f"{number} is odd")
```

```
65 is odd
37 is odd
76 is even
89 is odd
35 is odd
```

In [13]:
```
count=0
for i in range(5):
    name=input()            # 'python'
    if name=='python':      # 'python'=='python'    Tr
        count=count+1       # count=1+1= 2
```

```
python
python
p
p
p
```

In [14]:
```
count
```

Out[14]: 2

In [16]:
```
import random

even_count=0
odd_count=0

for i in range(5):
    number = random.randint(1,100)
    if(number %2==0):
        even_count=even_count+1    #  count=count+1     count+=1
        #even_count+=1
        print(f"{number} is even")
    else:
        odd_count=odd_count+1
        #odd_count+=1
        print(f"{number} is odd")

print("the number of eve numbers are:",even_count)
print("the number of odd numbers are:",odd_count)
```

```
98 is even
43 is odd
56 is even
84 is even
97 is odd
the number of eve numbers are: 13
the number of odd numbers are: 12
```

In [ ]:
```
#wap
#    ask the user get a random number (1,10), that as a number1
#    ask the user enter a number from keyboard, that as a number2
#    if number1==number2: print ( won)
#    else:  print(lost)

#   apply this 5 times which means give 5 chances
```

In [38]:
```
# Case-1:

for i in range(5):
    num1=random.randint(1,10)
    print(num1)
    num2=eval(input("enter a number:"))
    if num1==num2:
        print("won")
    else:
        print("lost")
```

```
2
enter a number:2
won
5
enter a number:4
lost
4
enter a number:4
won
5
enter a number:5
won
1
enter a number:1
won
```

In [39]:
```
# Case-2: whenever you won , code should stop

for i in range(5):
    num1=random.randint(1,10)
    print(num1)
    num2=eval(input("enter a number:"))
    if num1==num2:
        print("won")
        break
    else:
        print("lost")
```

```
1
enter a number:1
won
```

In [41]:
```python
# case-3: after how many attempts you won
for i in range(5):
    num1=random.randint(1,10)
    print(num1)
    num2=eval(input("enter a number:"))
    if num1==num2:
        print(f"you won after {i+1} attempt")
        break
    else:
        print("lost")
```

```
2
enter a number:3
lost
1
enter a number:1
you won after 2 attempt
```

In [36]:
```python
# case-4: how many chances left after loss
#num=5
chances=5
for i in range(chances):
    num1=random.randint(1,10)
    #print(num1)
    num2=eval(input("enter a number:"))
    if num1==num2:
        print(f"you won after {i+1} attempt")
        break
    else:
        print(f"you lost the attemps left is {chances-1-i}")

# 5-i     first   i=0   5-1-0=4
#         second  i=1   5-1-1=3
#         third   i=2   5-1-2=2
```

```
6
enter a number:7
you lost the attems left is 4
6
enter a number:7
you lost the attems left is 3
6
enter a number:7
you lost the attems left is 2
4
enter a number:7
you lost the attems left is 1
10
enter a number:7
you lost the attems left is 0
```

In [43]:
```python
# case-5: all the chances are over better luck next time

chances=5
for i in range(chances):
    num1=random.randint(1,10)
    #print(num1)
    num2=eval(input("enter a number:"))

    if num1==num2:
        print(f"you won after {i+1} attempt")
        break

    elif (chances-1-i)==0:
        print("all the chances are gone!")

    else:
        print(f"you lost the attemps left is {chances-1-i}")
```

```
enter a number:1
you lost the attemps left is 4
enter a number:2
you lost the attemps left is 3
enter a number:3
you lost the attemps left is 2
enter a number:4
you lost the attemps left is 1
enter a number:5
all the chances are gone!
```

In [ ]:
```python
# after all the chances are over better luck next time
num=5
for i in range(num):
    num1=random.randint(1,10)
    print(num1)
    num2=eval(input("enter a number:"))
    if num1==num2:
        print(f"you won after {i+1} attempt")
        break
    else:
        print(f"you lost the attems left is {num-1-i}")
```

In [28]:
```python
def win_loss():
    for i in range(5):
        num1=random.randint(1,100)
        num2=eval(input("enter the number :"))

        print(f"evaluate of {i} time")
        if num1==num2:
            print("you won")
            break
        else:
            print("you loss/ all the channes are lost")
win_loss()
```

```
enter the number :1
evaluate of 0 time
you loss/ all the channes are lost
enter the number :6
evaluate of 1 time
you loss/ all the channes are lost
enter the number :8
evaluate of 2 time
you loss/ all the channes are lost
enter the number :9
evaluate of 3 time
you loss/ all the channes are lost
enter the number :4
evaluate of 4 time
you loss/ all the channes are lost
```

In [29]:
```python
for i in range(5):
    num1=random.randint(1,10)
    print(num1)
    num2=eval(input("enter a number:"))
    if num1==num2:
        print("you won")
        break

    else:
        print("you lost")
print("all chances are over,better luck next time")
```

```
2
enter a number:2
you won
all chances are over,better luck next time
```

In [ ]:
```python
# wap ask the user to get sum of first 10 natural numbers

# 1  to 10
# 1+2+3+4+5+6+7+8+9+10=55
```

In [1]:
```
count=0

for i in range(5):
    count= count+1

count
# iteration-1:  c=0   new count= 0+1=1
# iteration-2:  c=1   new count=1+1=2
# i3            c=2   nc=2+1=3
# i4            c=3   nc=3+1=4
# i5            c=4   nc=4+1=5
```

Out[1]: 5

In [ ]:
```
0+1=1
1+2=3
3+3=6
6+4=10
10+5=15
15+6=21
21+7=28
28+8=36
36+9=45
45+10=55

count=count+1
count=count+i
```

In [2]:
```
summ=0
for i in range(1,11):
    summ=summ+i

print(summ)
```

```
55
```

In [3]:
```
# wap ask the user find the all divisiors of 10
# 10 is divided which numbers
# 10/1   10/2   10/5  10/10

# divisors of  10   max check iterations 10
# divisors of 20   max check iterations  20
# iterate the loop 10 times
# divided the same number with 1 to 10
# if the remiander ==0  : count=count+1


num=eval(input("which divisors you want:"))
count=0
for i in range(1,num+1):
    if num%i==0:
        count=count+1

print(f"the number of divsiors for {num} is {count}")
```

```
which divisors you want:10
the number of divsiors for 10 is 4
```

In [ ]: